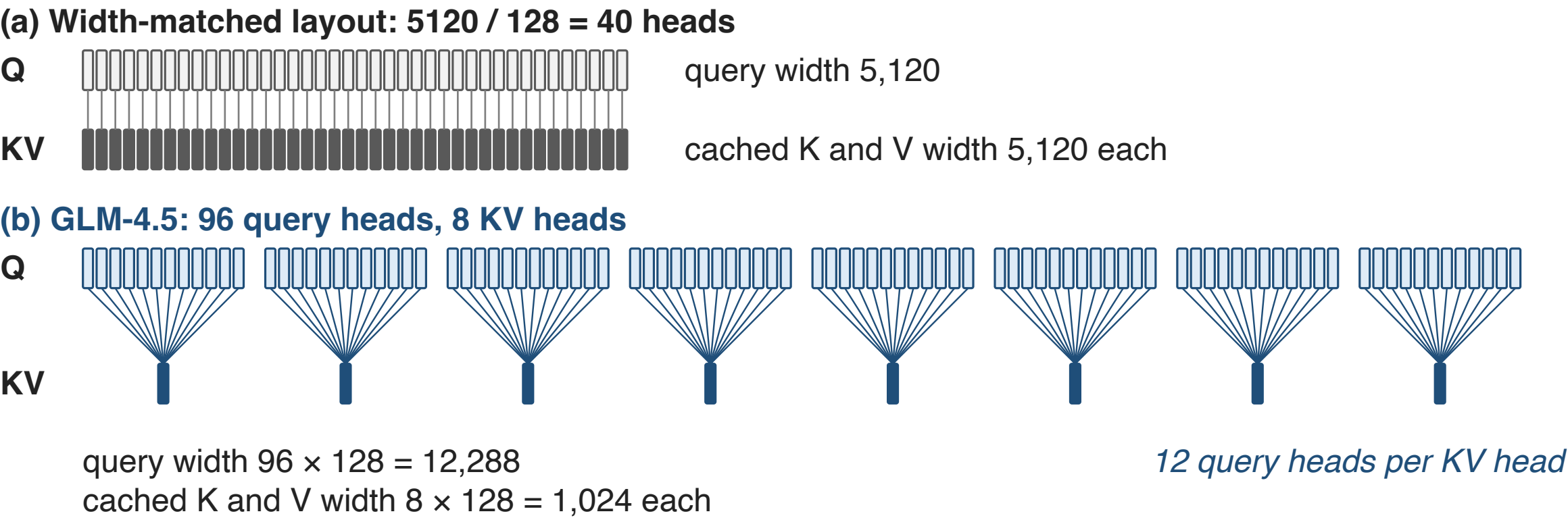

(a) Width-matched layout: 5120 / 128 = 40 heads
Q
KV
query width 5,120
cached K and V width 5,120 each
(b) GLM-4.5: 96 query heads, 8 KV heads
Q
KV
query width 96 × 128 = 12,288
cached K and V width 8 × 128 = 1,024 each
12 query heads per KV head